

ESP32 Multimodal NIDS — Progress Report

2026.08.13 · Matched-load evaluation · Board deploy · Domain robustness

Summary

Finding	Evidence
Training ↔ device density aligned	Firmware <code>win_pkts</code> / <code>win_dens</code> in windows
Matched-load dataset ready	6354 windows · balance OK · 3 attack types
Multimodal HIDS still matters	Dual-expert recovers stress-shaped attacks
Shallow full tree ≠ deployable	High FPR via host counters (<code>udpfail</code>)
Deploy choice: count-based wireless tree	Quiet-lab IDLE OK · DEAUTH fires
Absolute density is environment-sensitive	Busy air raises IDLE false alarms
Relative IDLE baseline helps	FPR 0.43→0.003 (matched busy NORMAL)
AUTH_FLOOD not ready as 4th class	Labels OK · <code>auth_packets</code> almost flat

HIPS remains off until false-positive behavior is stable across environments.

System (unchanged architecture)

Edge window IDS on ESP32 — not a per-packet firewall.

Stage	What
Sense	Promiscuous 802.11 + host counters
Decide	Shallow decision tree every 100 ms
Act	Host isolation disabled for now

Modality	In data?	Role now
Wireless counts / WIDS flags	Yes	Primary on the board tree
RF soft (RSSI / SNR)	Yes	Neutralized (domain shift)
HIDS (heap, queues, ...)	Yes	Neutralized on board · kept in analysis

Research remains multimodal; the board tree is a stable subset for demonstration.

Experiments this cycle

Stage	Dataset / setup	Goal
Density contract	Short busy capture	Train windows match board counts
Matched-load	20260809_000851	Busy NORMAL (label 0) + DEAUTH / SYN / ARP
Ablations	Same windows	full · wireless+RF · counts-only
Dual-expert	Same windows	Wireless v gated HIDS
Board check	Counts-only tree	IDLE / DEAUTH
Domain check (08.10)	Quiet vs busy air	Absolute vs relative density
AUTH_FLOOD probe	Two short labeled runs	Fourth attack feasibility

Dataset (matched-load)

6354 windows · 2026-08-09 · ~32 min

Class	Windows	Share
NORMAL (incl. busy)	4947	77.9%
DEAUTH	656	10.3%
SYN_FLOOD	495	7.8%
ARP_SPOOF	256	4.0%

Balance gate: **PASS**. Feature contract: firmware window counts (not syslog row counts).

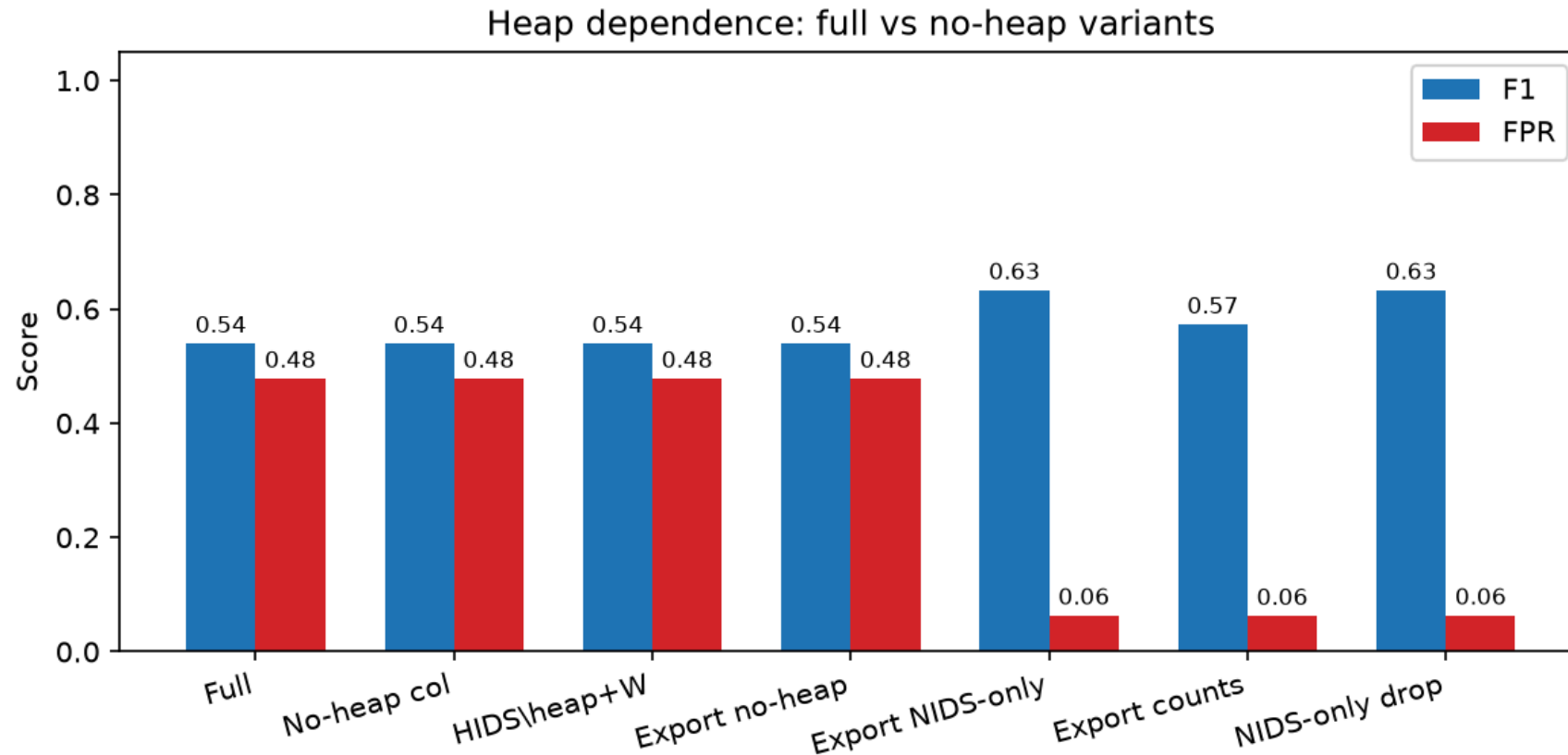
Main results (hold-out)

Model	F1	FPR	Notes
Full multimodal DT (depth 4)	0.54	0.48	Dominated by host pressure (<code>udpfail</code>)
Wireless + RF (no HIDS)	0.63	0.06	Fails on device (RSSI domain shift)
Counts-only (board)	0.57	0.06	IDLE OK · DEAUTH OK (quiet lab)
Random Forest	0.71	0.20	Stronger ensemble baseline
XGBoost	0.75	0.02	ARP recall still ~0.04
LightGBM	0.76	0.01	Best F1 here
Deeper DT (grid)	~0.72	—	Board keeps depth 4 for interpretability

Counts-only per-attack recall: DEAUTH ≈ 0.80 · SYN ≈ 0.37 · ARP ≈ 0.02

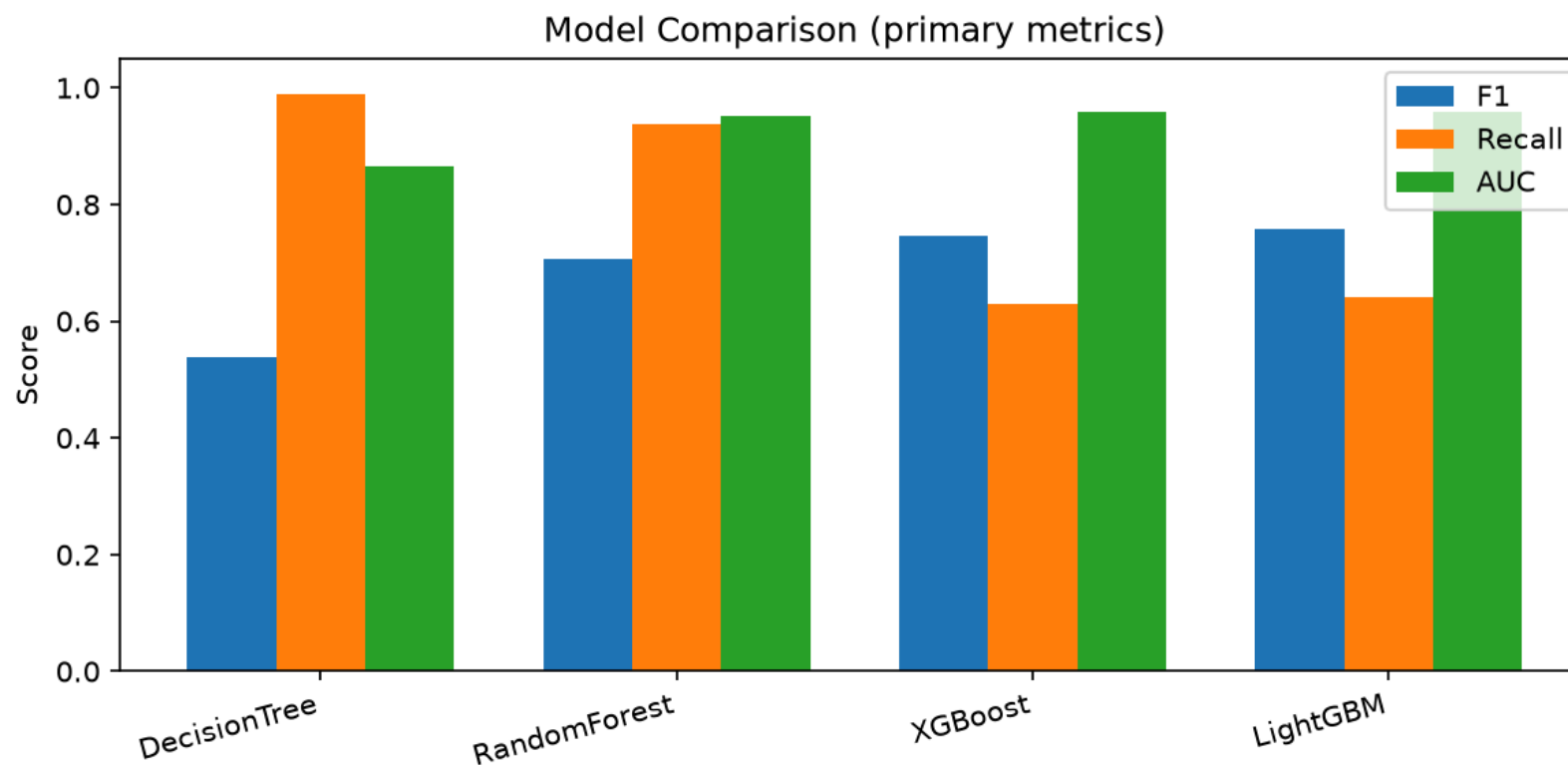
Stronger models improve F1/FPR but still miss ARP visibility — feature work remains the bottleneck.

Ablation (matched-load)



Matched-load reduced heap dominance vs Phase A; the full tree is still not deployable because other HIDS counters take over.

Model comparison



DT / RF / XGBoost / LightGBM on the same matched hold-out. Board demo uses shallow counts-only DT.

Why the board tree neutralizes HIDS + RF

Observation	Implication
Full tree FPR ≈ 0.48	Host counters track load , not attack identity
Wireless+RF looks good in hold-out	RSSI/SNR do not transfer → constant alarms on device
Counts-only quiet-lab smoke	DEAUTH path works; L2 attacks stay weak by design

Keep full multimodal + dual-expert for analysis; run counts-only on the board until fusion / better HIDS features are ready.

Dual-expert — idea

Two shallow trees, then combine:

Expert	Sees	Strength	Alone
W	Counts-only wireless	DEAUTH · low FPR	Misses stress-shaped SYN/ARP
H	HIDS only	SYN/ARP-shaped load	High FPR if always on

Rule	Behavior
Naive OR	Alarm if either fires → FPR explodes
Gated OR	Trust W always; add H only if $P(\text{attack} \text{H}) \geq 0.85$

Purpose: bring HIDS back as a **second path** without deleting the multimodal story.

Dual-expert — numbers (matched-load)

Fusion	F1	FPR	ARP	SYN	DEAUTH
W only (board today)	0.57	0.06	0.02	0.37	0.80
H only	0.60	0.35	1.00	0.92	0.94
W v H (naive)	0.57	0.40	1.00	0.94	0.96
W v H (gated)	0.63	0.09	0.02	0.49	0.96

- Gated OR: better F1 / FPR than naive OR; DEAUTH & SYN improve.
- **ARP still ≈ 0 → need air-visible ARP features, not a smarter OR.**

Board behavior & domain shift

Setting	IDLE	DEAUTH
Quiet lab	Mostly clear	Fires (deauth / targeted)
Busy RF environment (08.10)	Many false alarms	Still works when deauth present

Counts-only tree uses an **absolute** packet-count split (≈ 27 packets / 100 ms).
Background traffic above that threshold looks like “attack” even with label 0.

Environment	IDLE packet mean (window)
Quiet probe	~4
Busy probe	~31
Matched NORMAL (incl. busy)	~189

Relative IDLE baseline

Compare fixed threshold vs per-capture IDLE calibration ($\text{tot} > 2 \times \text{IDLE p90}$, or deauth present):

Dataset	Absolute FPR	Relative FPR	DEAUTH recall
Matched (busy NORMAL)	0.43	0.003	0.79 → 0.68
Busy air probe	0.52	0.12	—
Quiet probe	0.04	0.04	~0.85

Takeaway: quiet-lab pass $\neq$ robust deploy. Next step = **IDLE baseline on the board**, not only a deeper tree.

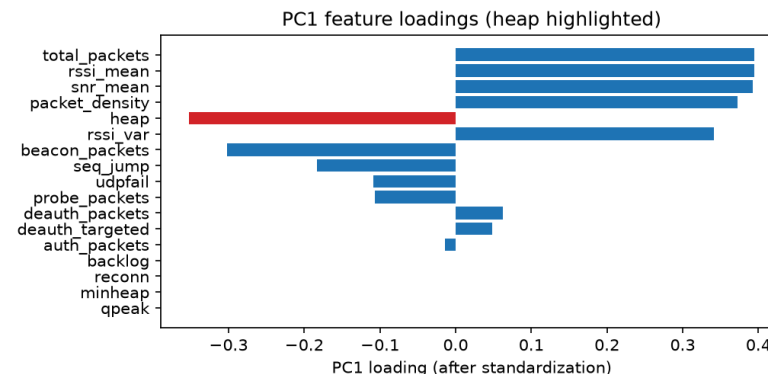
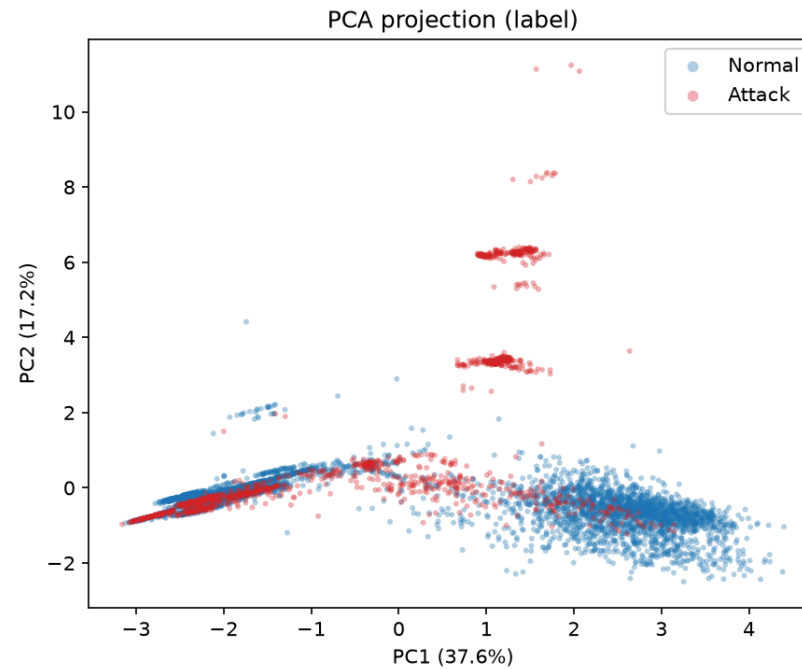
AUTH_FLOOD (feasibility)

Two short labeled runs (08.10):

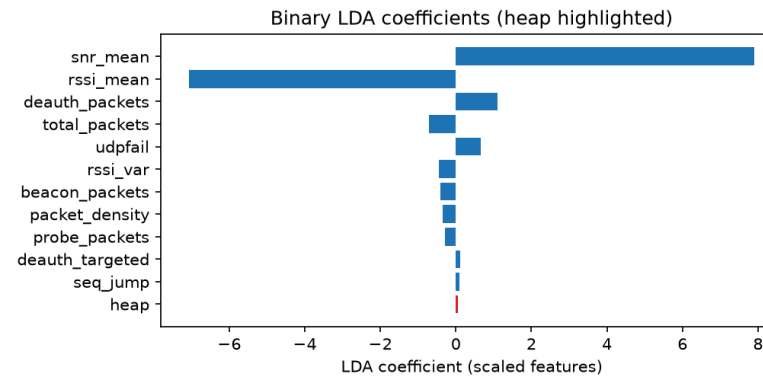
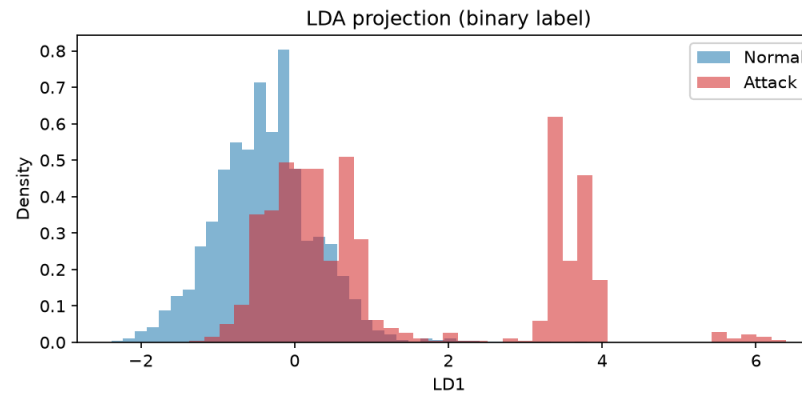
Check	Result
Collector START/STOP	OK
<code>aireplay</code> auth/assoc	Can succeed
Window <code>auth_packets</code>	$\approx 0-1$ · rarely non-zero
Board lights during AUTH	Unreliable (busy day $\approx$ density FP)

Conclusion: not ready as a fourth training class until auth (or association) counts are clearly visible on the ESP32. Prefer a stable AP later; optional teammate probe without Pi via Windows collector.

PCA (matched-load)

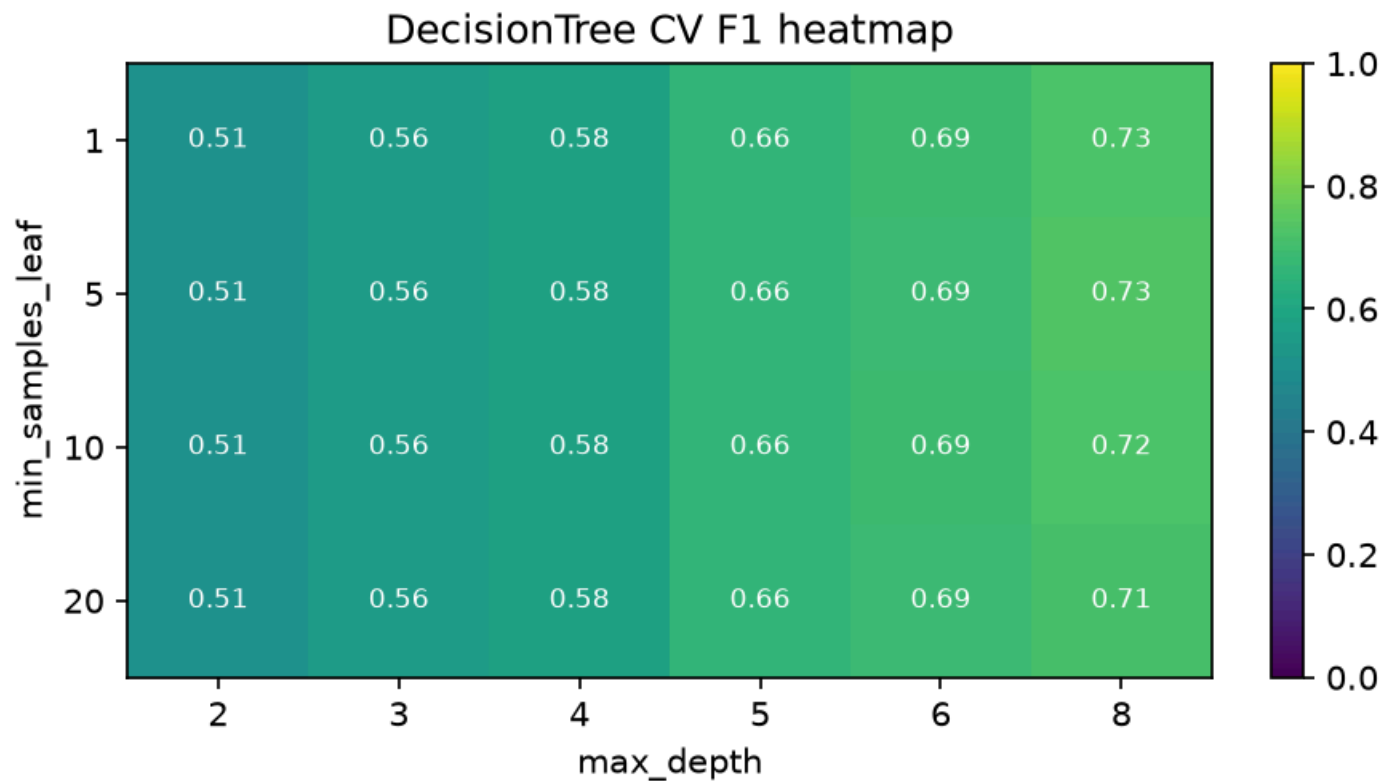


LDA (matched-load)



Supervised Normal/Attack axis; top coeffs include SNR/RSSI and deauth — RF directions overfit environments.

Hyperparameter grid (DT)



Deeper trees raise F1 (~ 0.72) · board tree stays **depth = 4** for interpretability — not because flash is full (~ 4 KB model).

Next steps

Priority	Work	Intent
1	ARP air-side features + recollect	Fix ARP invisibility (dual-expert cannot)
2	IDLE baseline calibration	Domain-robust deploy
3	AUTH_FLOOD revisit	After visible auth counts / stable AP
—	HIPS on · board-side ensemble	Deferred

Optional pre-meeting: short quiet IDLE → DEAUTH contrast capture (documentation only).

Ask

1. Proceed with **ARP** air-feature work next (after this meeting)?
2. Treat relative **IDLE** calibration as the **deploy robustness track** (before rushing AUTH)?
3. Defer **AUTH_FLOOD** until features are observable on device?

Recommendation: keep counts-only on the board · gated dual-expert as the HIDS path ·
prioritize **ARP** features + **IDLE** calibration · AUTH later.

Appendix — lab topology

```
Kali  -- air --> ESP32 (STA + promiscuous)
Kali  -- START/STOP --> Collector
ESP32 -- syslog (window stats) --> Collector → train → model.h
```

Attacks: deauth · SYN flood · ARP spoof · (AUTH exploratory) · 100 ms windows.

Appendix — Busy NORMAL (why)

Problem: Phase A NORMAL was mostly idle → tree learned “busy / low heap $\approx$ attack.”
Unlabeled load then looked like ~**98%** attacks.

Busy NORMAL = intentional background load with **label = 0**
(no attack START/STOP — teach “busy $\neq$ attack”).

Appendix — Busy NORMAL (how + result)

Segment	Label	Role
Quiet IDLE	0	True idle
Busy NORMAL	0	Moderate SYN-shaped load, unlabeled
DEAUTH / SYN / ARP	1	Scripted attacks with START → STOP

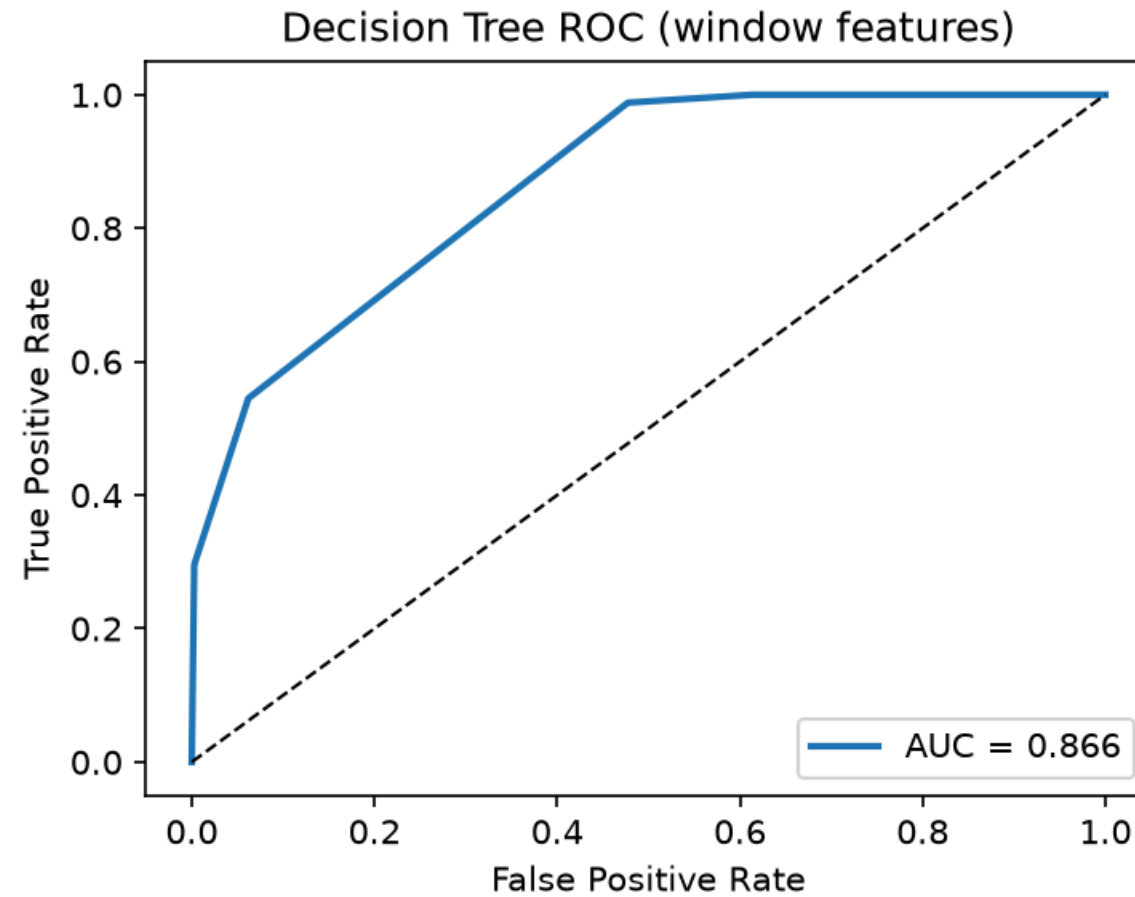
- Same file: load under **both** 0 and 1 → heap alone should not dominate.
- Result: busy NORMAL FP ~**10%** (vs ~98% on old tree).
- ≠ **busy RF air** (08.10): uncontrolled traffic → density FPs → motivates IDLE calibration.

Appendix — feature groups

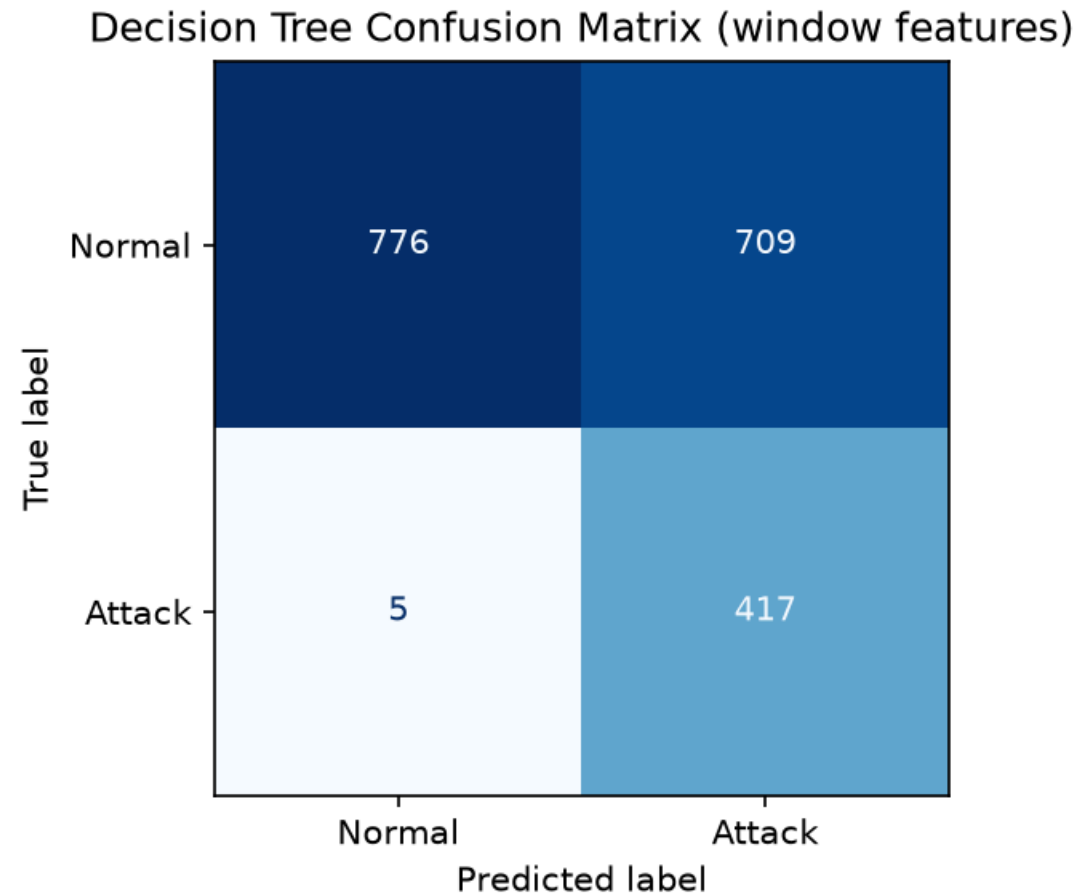
Group	Features
Counts / WIDS	total_packets , packet_density , beacon/deauth/probe/auth, deauth_targeted , seq_jump
RF soft	rss_i_mean , rss_i_var , snr_mean
HIDS	heap , minheap , reconn , qpeak , udpfail , backlog

Counts-only training: HIDS + RF soft filled with NORMAL median → splits on wireless counts.

Appendix — ROC

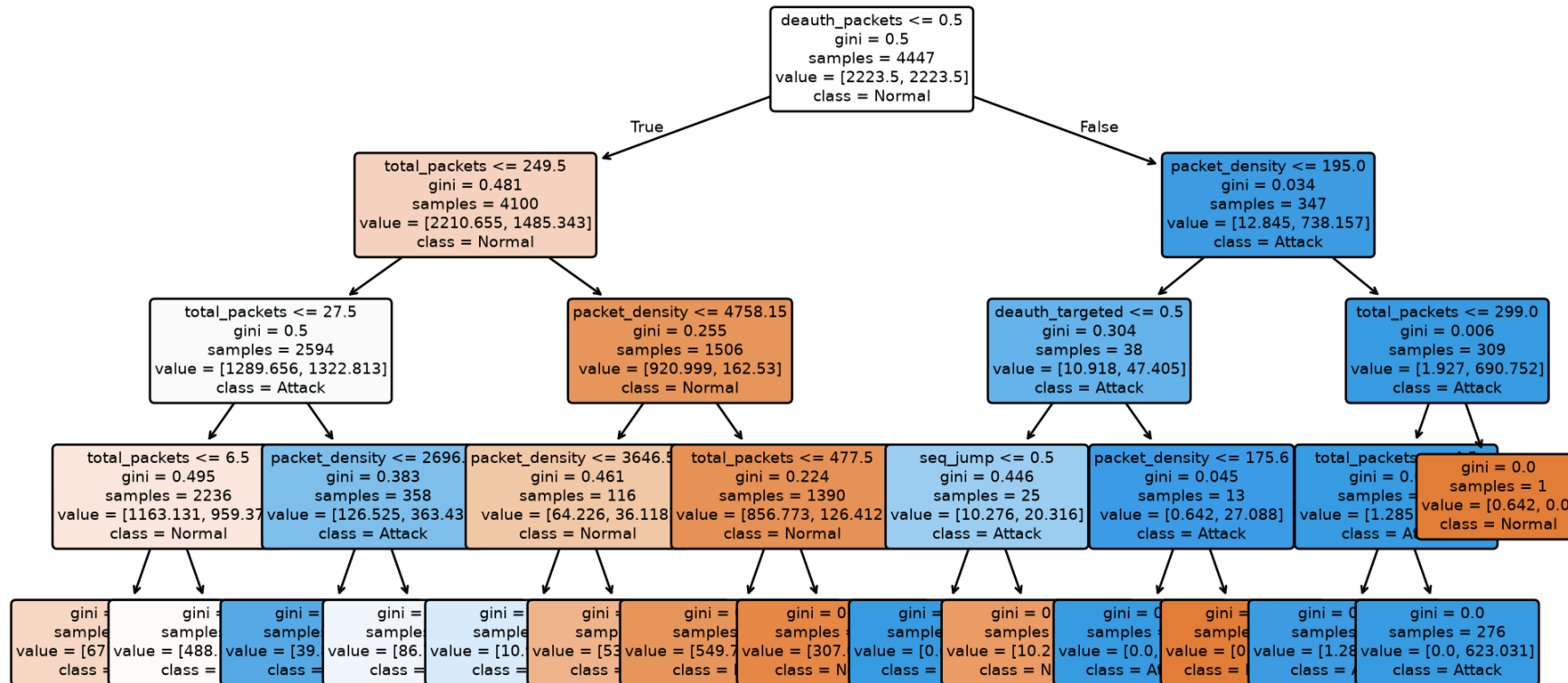


Appendix — Confusion matrix



Appendix — decision tree

Exported Decision Tree (DecisionTree+nids-counts)



Appendix — PCA variance

